Project Name: "Slam Dunk! Predicting the Total Points Scored in an NBA Game"
OIT 367 – Business Intelligence from Big Data
Final Prediction Project

Stanford Graduate School of Business
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Professor Mohsen Bayati

Group members:
Brad Allen
Chris Calway
Karen Chan
Russell Field
Walid Lassoued

Note: Will you let us keep your project? Yes.

Executive Summary:

Our project was to create a model that can predict the results of future NBA games. Specifically, we were asked by our project sponsors to create a model that can predict the total points scored in a game with greater accuracy than the programs used to generate the "over-under" betting line. We then refined the goal of the project even further – instead of generating a score prediction as the final product, we attempted to build a model that would provide a recommendation about whether to bet "over" the line, "under" the line, or not to bet at all given a certain confidence threshold. To create our model we were provided with a dataset consisting of ~46,000 rows of 66 columns, with each row representing one game from the perspective of one team.

The result of our work is a linear model that uses approximately 100 numeric and categorical variables. When provided with a confidence threshold, the model can determine whether the user should bet "over", "under", or not at all.

To create this model we performed several cleaning and pre-processing steps, and these were primarily done in Excel. Since each row of the data contained primarily in-game statistics whereas bets must be placed prior to the start of a game, much of the pre-processing was focused on creating new columns that reflected information known at the start of the game (e.g. the number of points scored by the home team in the last game they played). Although the data was quite clean and consistent, some additional processing was required to change blanks and NAs to zeroes or to the mean value of the predictor depending on the context (described in greater detail below). Categorical variables were binned prior to development of the models – the main variables binned were month, day, time zone, division, conference, and playoffs.

Features and predictors were generated by prior knowledge of basketball, by visual analysis of the data, and by examining correlation matrices.

We generated four models using variations of linear and logistics regressions, and we evaluated these models based on RMSE (for the linear models) and on AUC (for all four). We eventually found that the linear regressions yielded superior results. We explored interaction terms using BD Logistic Regression, but the AUC for this model was worse than that generated by the linear model, and few interaction terms showed to have high predictive power made intuitive sense. Although the AUC of our models never achieved a very high level (our highest AUC was 0.517), we believe that this is a reasonable result given the tremendous amount of time and resources that professionals have devoted to the creation of models supporting over-under lines.

The primary conclusion of our report is that professionally produced over-under lines are sophisticated and robust, and they cannot easily be outplayed even when considering a large array of potential predictors. The potential exists to make a small amount of money, but very few games pass the required confidence threshold, and thus the process would be very slow.

Problem Statement:

The purpose of our project was to create a model that can predict the results of future NBA games. Specifically, we were tasked by our project sponsors with creating a model that can predict the total points scored in a game with greater accuracy than the programs used to generate the "over-under" betting line. As our project progressed we refined this further – instead of generating a score prediction as the final product, we wanted a model that would provide a recommendation about whether to bet "over" the line, "under" the line, or not to bet at all given a certain confidence threshold.

Analysis:

1 - Data Pre-Processing:

The data we used was provided to us by our sponsors, and they collected it primarily from http://sportsdatabase.com/nba/query. The raw data consists of ~46,000 rows of 66 columns contained in a .CSV file (total size ~35MB), with each row representing one game from the perspective of one team. For example, the first row of the raw data represents a game played on Nov 3, 1995 from the perspective of the Bulls (playing against the Pelicans), and it contains data about attendance at the stadium, number of free throws attempted by the Bulls, number of free throws made by the Bulls, number of fouls committed by the Bulls, etc. The second row represents the same game (Nov 3, 1995), but now from the perspective of the Pelicans (playing against the Bulls), so all of the data in this row represents information about the Pelicans (number of free throws attempted, number of free throws made, number of fouls committed), as well as some general information that is duplicated (attendance at the stadium). The full data dictionary explaining all 66 columns is contained in the technical appendix.

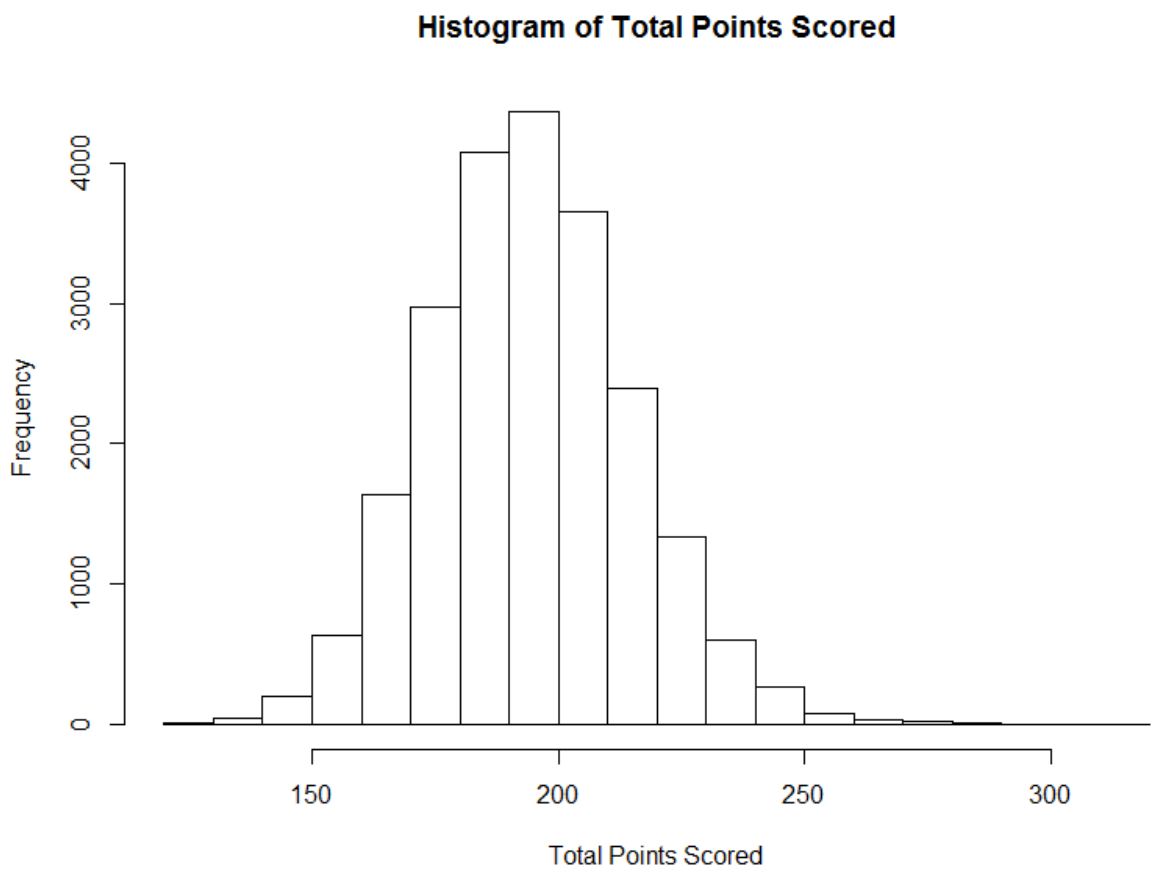
Our first pre-processing step was to merge each pair of rows into a single row that contained all of the information for an individual game. For example, keeping with our example of the game on Nov 3, 1995, the new pre-processed data set represented that game as a single row, and all of the team specific data was folded into new columns like "Home Field Goals Attempted" and "Away Fouls Committed" with the "Home" team clearly identified as the Bulls and the "Away" team clearly identified as the Pelicans. All of this pre-processing was performed in Excel.

After this initial pre-processing step, we examined each column to determine where values were missing. The data were remarkably clean and consistent, but there were some columns where values were not tracked prior to the start of a particular year (e.g. all of the "playoff statistics" – series game, series games, series losses, series wins, and round – were not tracked before 2002, so all values from 1995 to 2001 were blank). For the columns where missing data were grouped together (as in the case of playoff data), we replaced the missing values with

zeroes. Where missing data were interspersed in a column that was otherwise complete, we replaced the missing value with the mean of the distribution. In both cases we added a dummy variable to note which values were originally missing so that we could later refer to these values and potentially create interactions terms.

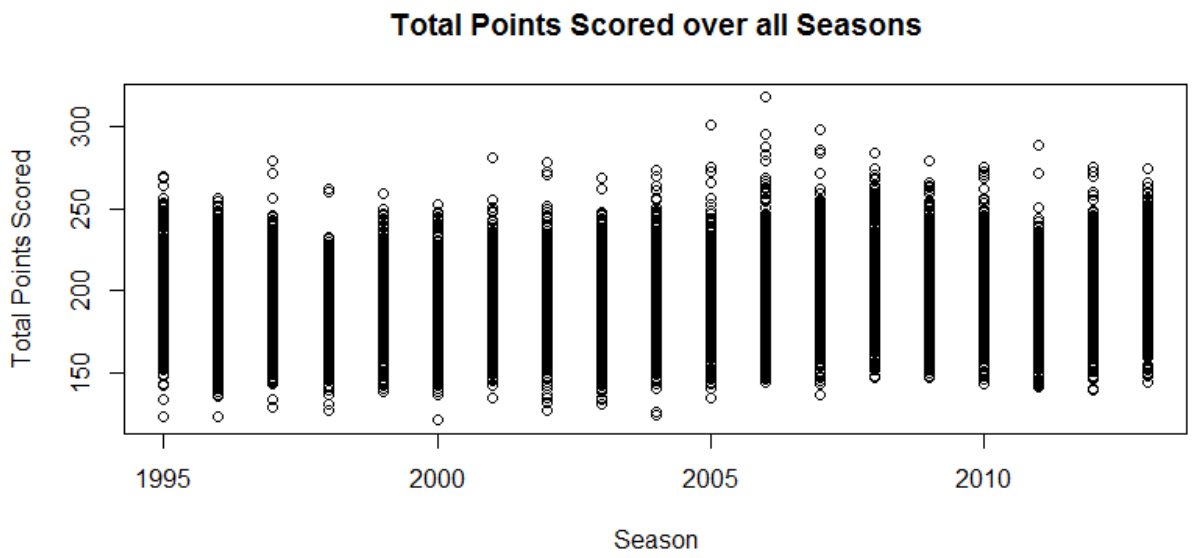
2 – Feature Engineering:

After the initial pre-processing of the data, our first step was simply to examine the primary response variable (total score) and determine if we could see any major trends.



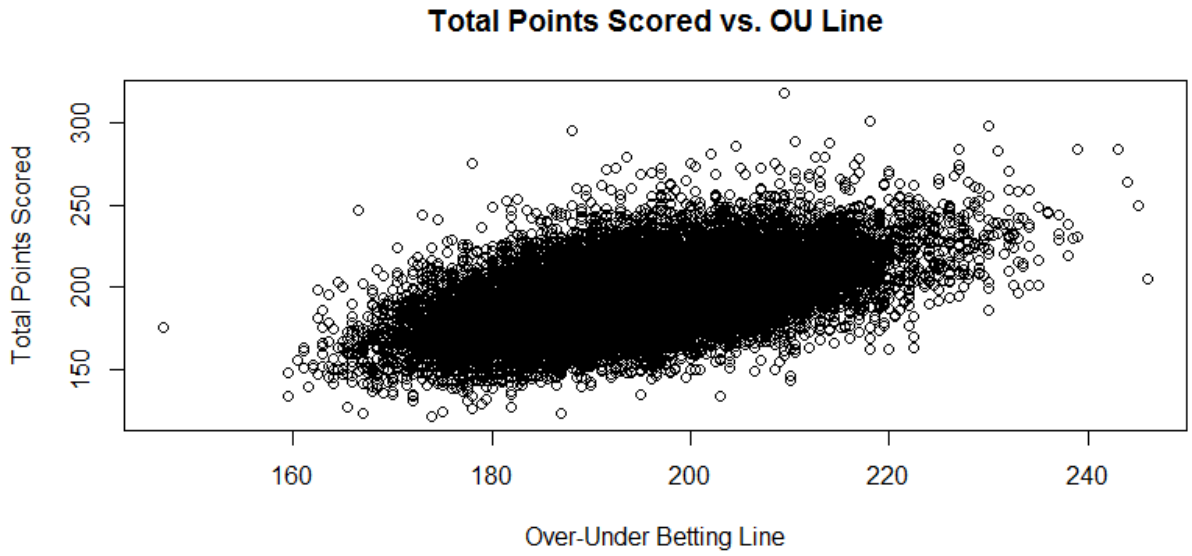
As seen above, the distribution of total scores appears nearly normally distributed, although the right tail is substantially longer than the left.

Since our data is time series, we also wanted to see how the values for total points scored have changed over different seasons:

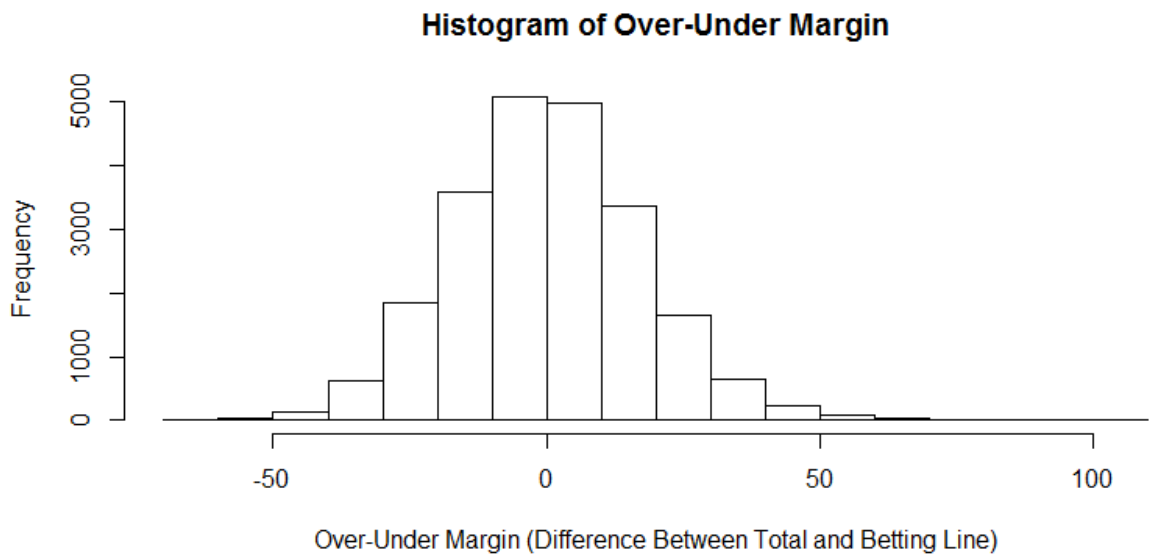


As seen above, it does not appear that there are any immediately obvious year-over-year trends in total points scored.

Our last visualizations before we started building features were to get a sense of the benchmark that we were working against – the over-under betting line. To this end, we looked at a scatterplot of total points scored vs. the over-under betting line, and a histogram of the OU margin (the difference between the total score and the line).



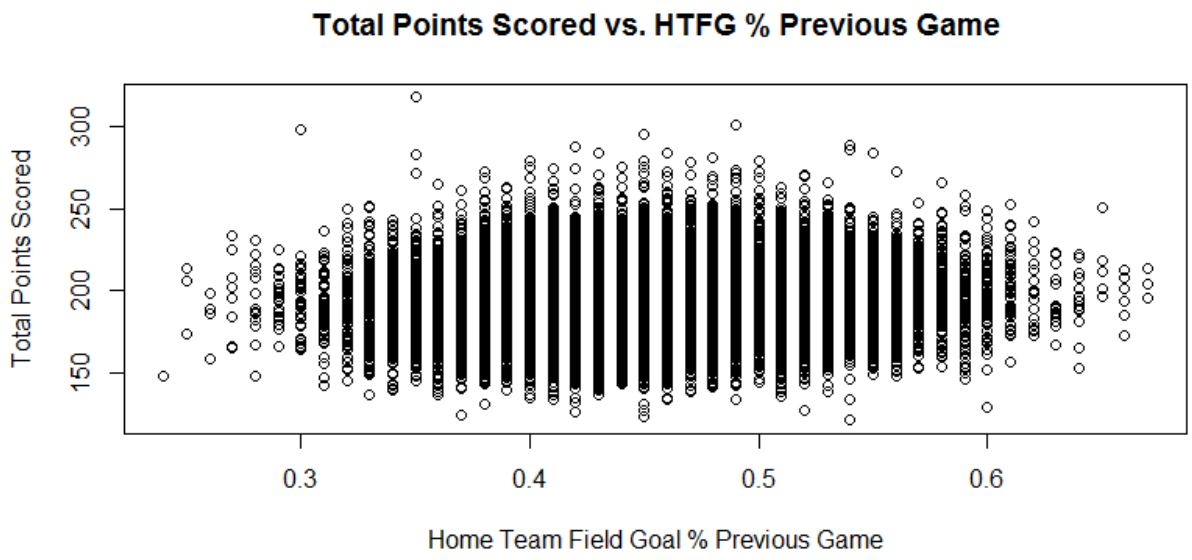
As expected, there is a clear positive correlation between total points scored and the OU margin, but there is also significant noise.

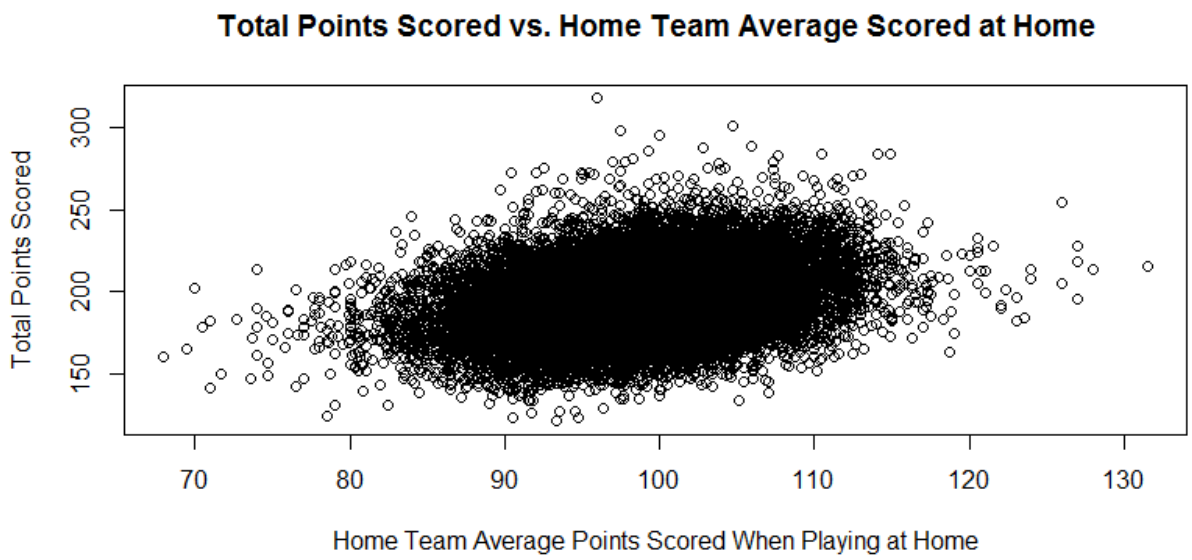


Looking at the over-under margin, the distribution conforms extremely well to the normal (as expected), but the peak is very slightly to the left of zero, and the right tail is longer than the left. The long right tail matches up with the distribution of total points scored, and it suggests that the model used to generate the betting line struggles to correctly predict blowout games with total scores much higher than average.

With this information in mind, we worked to determine which factors might contain substantial prediction power about the total score of a particular game. Since the goal of our project was to create a model that could provide predictions *before* a game began, we knew that we were limited to information that would be available prior to the start of play. Since each row of our pre-processed data still contained mostly in-game statistics, we began creating new columns that would capture information about prior performance. For example, in addition to recording the field goals scored by the home team in a particular row, we also began recording the field goals that were scored by the home team in the previous game, and in the game before that. All of this data manipulation was done in Excel.

We began by using our intuition and knowledge of basketball to create simple columns that seemed likely to influence total score. For example, since teams with greater shooting efficiency (field goals made divided by field goals attempted) are likely to score more points, it seemed reasonable that a team's average shooting efficiency for all games prior to the current game is likely a strong predictor of that team's score, and by extension the total score of the current game. After creating these simple predictors, we created scatter plots to get a visual sense of the relationships.





As seen above, some of our early hypotheses about “simple” and “obvious” relationships ended up being completely false, as the data above show few meaningful trends (there is a slight positive correlation between Average Points Scored by Home Team at Home and Total Points Scored).

We continued to generate potential predictor columns based on questions that arose within our group (e.g. does game attendance influence scoring?) as well as suggestions from our project sponsors (e.g. does the day of the week have an impact on total points?), and we did an initial trial of these predictors by creating a correlation matrix and examining the coefficients. Using the Corrplot library, we created a heatmap of correlations (see Exhibit B), with larger, darker circles indicating stronger relationships (with blue as a positive correlation, and red as a negative correlation).

As seen in both versions of the heatmap, there are very few dark circles in the first line, with the only exception being the correlation between total points scored and itself, which is obviously 1. The notable predictors include the over-under betting line (the strongest correlation), average number of points scored by the home team, and average number of points scored by the away team. A full list of predictors sorted by correlation coefficient is available in the technical appendix as Exhibit C.

Of the variables considered in the model, we identified 6 categoricals that required binning. These variables included: month, day, time zone, conference, division, and playoffs. Binning categorical variables extended our predictors by 19 variables.

3 – Model Development

We explored 4 models for creating predictions as we refined our sense of the project's goals. We began by using a linear model to predict total score, and we trained this model using all available columns (except those which contained in-game data which would normally not be available prior). We also implemented a linear model with lasso using glmnet, and we evaluated the results of both linear models by comparing the RMSE of the total score prediction and the AUC of the prediction on whether to bet over (1) or under (0).

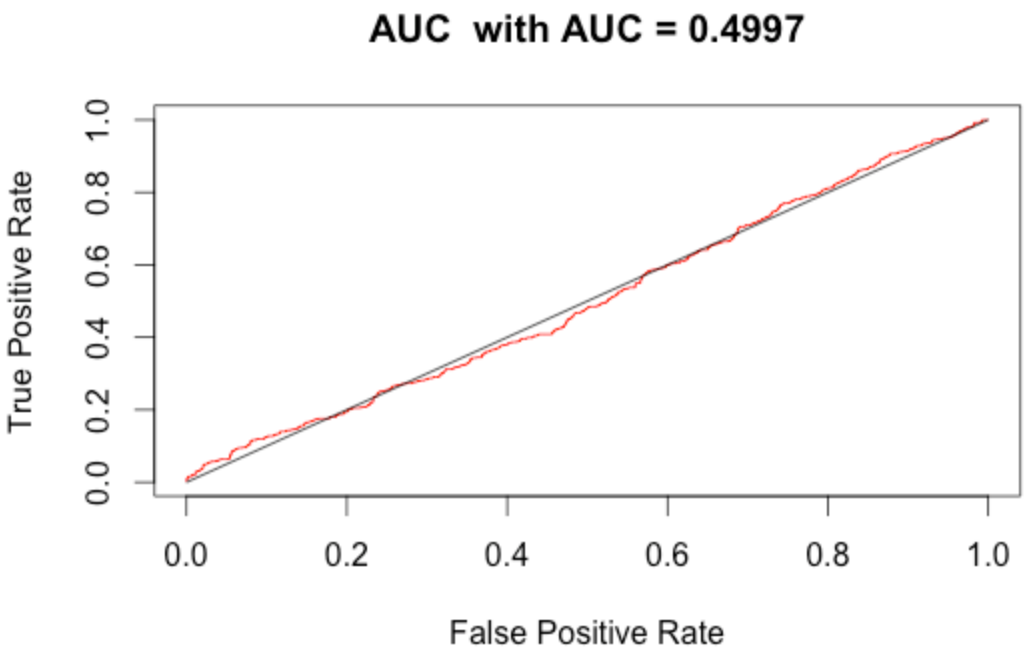
Next, we created a logistic regression model and compared the results for the AUC to the two linear models. As the AUC for the logistic regression was inferior to the two linear models, we tried to pursue a different route, and we implemented a BD Logistic Regression to determine whether any of the interaction terms between our variables were relevant predictors of total score. The results of this model were not encouraging – the AUC was lower than any of the other models we had considered, and the interaction terms with high predictive power were far from intuitive and seemed reflective of fringe effects with small numbers of observations.

With all of this in mind, we decided to focus our energy on the linear regressions. Although our data are time-series, the relevant information about previous performances are already included in each individual row, so we were able to randomize and implement cross-validation without distorting the underlying trends.

Principal Findings, Conclusions, and Recommendations:

The primary conclusion of our report is that professionally produced over-under lines are sophisticated and robust, and they cannot easily be outplayed even when considering a large array of potential predictors.

We were never able to produce impressive AUC values when our models were running correctly, and the highest that we were able to achieve was 0.5017. Below is a plot of our AUC for the linear model that we eventually settled on as our most robust and well-designed.



By altering the confidence threshold at which we were willing to bet (a bound around predicted total score divided by line) we could achieve relatively high success rates (>80% successful bets), but these came at the expense of very few games meeting the threshold criteria (<100 out of a set of 22,000). This suggests that the potential exists to make a small amount of money, but the process would be very slow, and it would require that the assumptions underlying the model not change over many years.

TECHNICAL APPENDICES:

EXHIBIT A: Data Dictionary

Source: http://sportsdatabase.com/nba/query

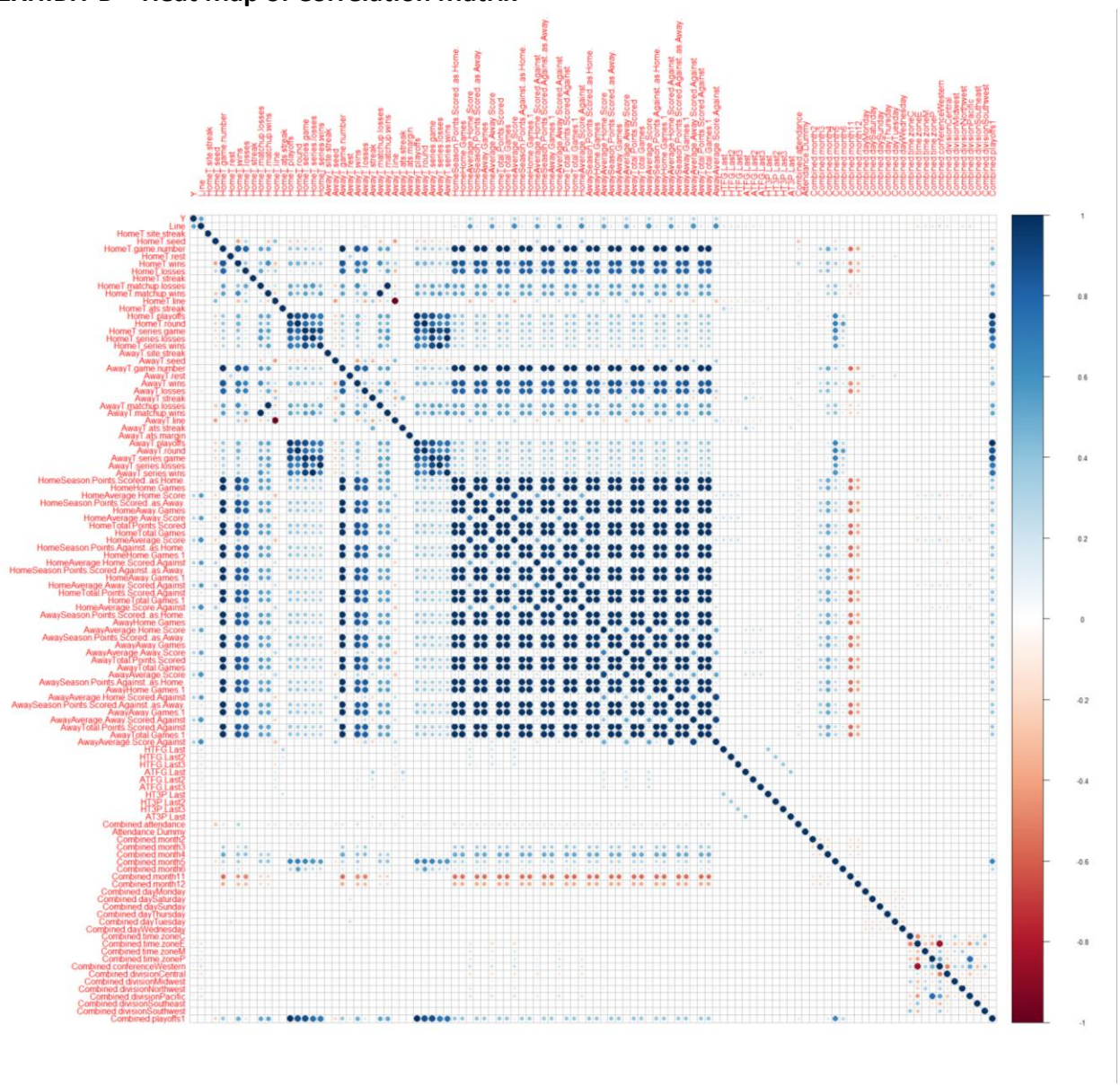
Most definitions located at: http://www.google.com/url?q=http%3A%2F%2Fkillersports.com%2FDownload%2FNBA%2Fquery_manual.pdf&sa=D&sntz=1&usg=AFQjCNH0o_EyQK25c67_U4NMBL5lQNx86g

Field	Description
date	Date of Game
assists	Total assists for one team
ats margin	"Against the Spread" (+ means performed better)
ats streak	Length of Sequence of +'s
attendance	Total attendance of Game
biggest lead	In-Game Lead
blocks	Total blocks for Team
conference	Conference of Game
day	Day of the Week for the Game
defensive rebounds	Total number of defensive rebounds
division	Conference Division
dpa	Defensive Points Added
dps	Delta Points Scored. Difference between points the team was expected to score and points actually scored.
fast break points	Points scored on a Fast Break
field goals attempted	
field goals made	
fouls	
free throws attempted	
free throws made	
game number	
line	
losses	
margin	
margin after the first	
margin after the third	
margin at the half	
matchup losses	Stats for that basketball season
matchup wins	Stats for that basketball season
minutes	
month	
offensive rebounds	
officials	Referees for the game
opponents	Teams played (so far in the season)
ou margin	The margin by which a game went over or under the vegas OU line
ou streak	
overtime	TRUE/FALSE - Did the game go into OT?
playoffs	0 or 1, depending on if it is a playoff game
points	
points in the paint	
quarter scores	
rebounds	
rest	Number of days off a team has between games. If a team plays on consecutive days, their rest is zero.
round	Playoff Statistic. Does not start getting counted until 2002.
season	
seed	Rank in Conference (from 1-8). Does not start until 1998. Starts at beginning of season.
series game	Playoff Statistic. Does not start getting counted until 2002.
series games	Playoff Statistic. Does not start getting counted until 2002.
series losses	Playoff Statistic. Does not start getting counted until 2002.
series wins	Playoff Statistic. Does not start getting counted until 2002.

EXHIBIT A: Data Dictionary

site	
site streak	Consecutive wins or losses (for both home and away)
steals	
streak	Consecutive wins (total)
team	
o:team	
team rebounds	
three pointers attempted	
three pointers made	
time of game	
time zone	
total	
turnovers	
wins	
official1	
official2	
official3	
game id	

EXHIBIT B – Heat Map of Correlation Matrix



The heatmap above is difficult to read given the large number of variables, so included below is a cropped version of the image showing only the first line (correlation between total points scored (shown as “Y”) and all other predictor variables).

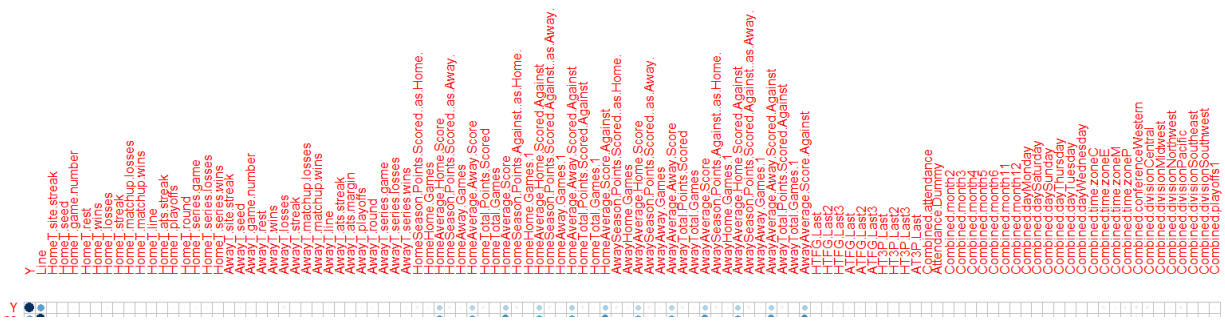


EXHIBIT C - Correlation Coefficients of Predictors with Total Score

Predictor	Correlation with Total Points Scored	Absolute Value of Correlation
Total Points Scored	1	1
Line	0.53622057	0.53622057
HomeAverage.Score.Against	0.338547817	0.338547817
AwayAverage.Score.Against	0.338306191	0.338306191
HomeAverage.Score	0.321892467	0.321892467
HomeAverage.Home.Scored.Against	0.316004713	0.316004713
AwayAverage.Away.Scored.Against	0.309882382	0.309882382
AwayAverage.Score	0.30797926	0.30797926
AwayAverage.Home.Scored.Against	0.30485025	0.30485025
HomeAverage.Away.Scored.Against	0.302230426	0.302230426
HomeAverage.Home.Score	0.30005777	0.30005777
HomeAverage.Away.Score	0.284629128	0.284629128
AwayAverage.Away.Score	0.282565962	0.282565962
AwayAverage.Home.Score	0.279201976	0.279201976
Combined.conferenceWestern	0.111968329	0.111968329
Combined.divisionPacific	0.104616424	0.104616424
Combined.time.zoneE	-0.096221537	0.096221537
Combined.divisionCentral	-0.096128876	0.096128876
HomeSeason.Points.Against..as.Home.	0.089138787	0.089138787
AwaySeason.Points.Scored.Against..as.Away.	0.088419902	0.088419902
HomeTotal.Points.Scored.Against	0.087921674	0.087921674
AwayTotal.Points.Scored.Against	0.087882189	0.087882189
AwaySeason.Points.Against..as.Home.	0.086181879	0.086181879
HomeSeason.Points.Scored.Against..as.Away.	0.08559729	0.08559729
Combined.time.zoneP	0.085323542	0.085323542
HomeSeason.Points.Scored..as.Home.	0.084454484	0.084454484
HomeTotal.Points.Scored	0.083481073	0.083481073
AwaySeason.Points.Scored..as.Away.	0.082831324	0.082831324
AwayTotal.Points.Scored	0.082290057	0.082290057
HomeSeason.Points.Scored..as.Away.	0.08137381	0.08137381
AwaySeason.Points.Scored..as.Home.	0.080611648	0.080611648
AwayT.losses	0.079194608	0.079194608
Combined.divisionNorthwest	0.078294089	0.078294089
Combined.time.zoneM	0.077454204	0.077454204
HTFG.Last	0.073171465	0.073171465
HomeT.losses	0.070910326	0.070910326
HTFG.Last3	0.066489709	0.066489709
AwayT.seed	0.063403727	0.063403727
ATFG.Last3	0.060080666	0.060080666
HTFG.Last2	0.059914005	0.059914005
ATFG.Last	0.056265753	0.056265753
HomeHome.Games	0.055704837	0.055704837
HomeHome.Games.1	0.055704837	0.055704837
HomeTotal.Games	0.055182759	0.055182759
HomeTotal.Games.1	0.055182759	0.055182759
HomeT.game.number	0.05518211	0.05518211
ATFG.Last2	0.05516538	0.05516538

EXHIBIT C - Correlation Coefficients of Predictors with Total Score

Predictor	Correlation with Total Points Scored	Absolute Value of Correlation
AwayTotal.Games	0.055159369	0.055159369
AwayTotal.Games.1	0.055159369	0.055159369
AwayT.game.number	0.055157401	0.055157401
AwayAway.Games	0.055090887	0.055090887
AwayAway.Games.1	0.055090887	0.055090887
AwayHome.Games	0.054525282	0.054525282
AwayHome.Games.1	0.054525282	0.054525282
HomeAway.Games	0.053970612	0.053970612
HomeAway.Games.1	0.053970612	0.053970612
HomeT.seed	0.050387262	0.050387262
HomeT.round	-0.043992724	0.043992724
AwayT.round	-0.043992724	0.043992724
Combined.divisionMidwest	-0.043544115	0.043544115
HomeT.playoffs	-0.042843578	0.042843578
AwayT.playoffs	-0.042843578	0.042843578
Combined.playoffs1	-0.042843578	0.042843578
HomeT.series.game	-0.042570605	0.042570605
AwayT.series.game	-0.042570605	0.042570605
HomeT.series.losses	-0.041452516	0.041452516
AwayT.series.wins	-0.041452516	0.041452516
Attendance.Dummy	-0.033318853	0.033318853
HomeT.series.wins	-0.032461214	0.032461214
AwayT.series.losses	-0.032461214	0.032461214
Combined.month6	-0.031693882	0.031693882
Combined.month11	-0.029936744	0.029936744
HT3P.Last3	0.024976756	0.024976756
Combined.month5	-0.024935711	0.024935711
HT3P.Last2	0.02478504	0.02478504
HomeT.wins	0.024094597	0.024094597
HT3P.Last	0.022978472	0.022978472
Combined.month3	0.020743626	0.020743626
HomeT.rest	0.019039553	0.019039553
HomeT.line	-0.018723737	0.018723737
AwayT.line	0.018723737	0.018723737
Combined.dayThursday	-0.017377104	0.017377104
Combined.month2	0.017163609	0.017163609
AwayT.wins	0.016782036	0.016782036
Combined.attendance	-0.01569307	0.01569307
Combined.dayWednesday	0.015151209	0.015151209
Combined.divisionSouthwest	0.01309922	0.01309922
AT3P.Last	0.012585222	0.012585222
AwayT.site.streak	-0.012266281	0.012266281
AwayT.ats.streak	-0.011946159	0.011946159
HomeT.ats.streak	0.010595563	0.010595563
AwayT.streak	-0.010160002	0.010160002
Combined.time.zoneC	-0.009708414	0.009708414
Combined.month4	0.008947661	0.008947661

EXHIBIT C - Correlation Coefficients of Predictors with Total Score

Predictor	Correlation with Total Points Scored	Absolute Value of Correlation
Combined.daySaturday	-0.008798608	0.008798608
AwayT.ats.margin	-0.00867436	0.00867436
Combined.daySunday	-0.008285095	0.008285095
Combined.month12	-0.008003579	0.008003579
Combined.divisionSoutheast	0.007400346	0.007400346
HomeT.matchup.losses	-0.005500732	0.005500732
AwayT.matchup.wins	-0.005500732	0.005500732
Combined.dayMonday	0.005241137	0.005241137
AwayT.rest	-0.004483939	0.004483939
HomeT.matchup.wins	-0.003363297	0.003363297
AwayT.matchup.losses	-0.003363297	0.003363297
HomeT.streak	-0.002111162	0.002111162
HomeT.site.streak	0.001348311	0.001348311
Combined.dayTuesday	2.79E-05	2.7937E-05

EXHIBIT D – R Code

```
rm(list=ls())
source("http://www.stanford.edu/~bayati/oit367/T367_utilities_11_beta.R")
#setwd("/Volumes/ccalway$/My Documents")
setwd("/Users/walid/Documents/OIT Big Data/NBA")

data = read.csv("NBAdata Calway v4.csv")

data =na.omit(data)
#data= data[data$OU.margin!=0,]
data3 <- data[,6:12]
data2 = data[,16:95]
Total = data[,2]
Line = data[,3]
data1= cbind(Total,Line,data2,data3)
head(data$total)
##BIN THE CATEGORICAL VARIABLES
names.categorical.predictors =
c("Combined.month","Combined.day","Combined.time.zone","Combined.conference"
                        ,"Combined.division","Combined.playoffs")
x = colnames(data1)
names.numerical.predictors = x[!(x %in% names.categorical.predictors)]
numerical.predictors = data1[,names.numerical.predictors]
categorical.predictors = data1[,names.categorical.predictors]
categorical.predictors.binned = bin.cat.data(categorical.predictors)
data.processed = cbind(numerical.predictors,categorical.predictors.binned)

#numerical.sparse = as(as.matrix(numerical.predictors),"sparseMatrix")

#data.processedx = cBind(numerical.sparse, categorical.predictors.binned)
A = data.processed[,2:6]
B = data.processed[,7:ncol(data.processed)]
C = data.processed[,1]
X = cbind(A,B)
X = sapply(data.frame(X),as.numeric)
Y = data.processed$Total
Y = sapply(data.frame(Y),as.numeric)
lmdata = cbind(Y,X)

#train_ind = 1:(.6*nrow(X))

# 21043 is the start of the 2013 season
train_ind = 1:21043
```

```
# 21679 is the mid-point of the 2013 season
valid_ind = 21043:nrow(X)

# We will train on all the seasons up to 2013, we discard half of the 2013 season and validate
# on the remaining of the 2013 season

lmdata_training = as.data.frame(lmdata[train_ind,])
#lmdata_validation = as.data.frame(lmdata[-train_ind,])
lmdata_validation = as.data.frame(lmdata[valid_ind,])

m2 = lm(Y~.,data=lmdata_training)
Ypred2 = as.data.frame(predict(m2, newdata=lmdata_validation))

cat('RMSE for LM is',rmse(Ypred2,Y[-train_ind]),'\n')

##--------------------

#predGame2 = Ypred2>Line[-train_ind]
#actualGame = lmdata_validation$Y>Line[-train_ind]
#mean(predGame2==actualGame)


predGame2 = ifelse(Ypred2/Line[valid_ind]>1.015,1,ifelse(Ypred2/Line[valid_ind]<0.98,0,NA))

actualGame = as.numeric(lmdata_validation$Y>Line[valid_ind])
head(actualGame)
strategy=cbind(predGame2,actualGame)
strategy =na.omit(strategy)

mean(strategy[,1]==strategy[,2])

####################################
# Calculating the AUC and plotting the curve
delta = as.numeric(as.matrix(Ypred2 - Line[valid_ind]))
probAboveLine = 1/(1+exp(-delta))
IsAboveLine = as.numeric(lmdata_validation$Y>Line[valid_ind])
auc(IsAboveLine,probAboveLine)
plot.roc(IsAboveLine,probAboveLine, main="AUC")
```